CCC-C-437a
MAY 28, 1958
SUPERSEDING
Int. Fed. Spec. CCC–C–00437(GSA–FSS)
October 7, 1957 and
Fed. Spec. CCC–B–686a
March 2, 1948

FEDERAL SPECIFICATION

# CLOTH, COTTON, BROADCLOTH, MERCERIZED

*This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.*

## 1. SCOPE AND CLASSIFICATION

**1.1 Scope.**—This specification covers the requirements for cloth, cotton, broadcloth.

**1.2 Classification.**

1.2.1 *Types.*—The cloth, cotton, broadcloth covered by this specification shall be of the following types (see 6.5):

Type I.—148 by 74, 3.5 ounces per square yard.
Type II.—140 by 56, 3.2 ounces per square yard.
Type III.—132 by 64, 3.2 ounces per square yard.
Type IV.—116 by 56, 3.2 ounces per square yard.
Type V.—102 by 56, 3.2 ounces per square yard.
Type VI.— 82 by 56, 2.8 ounces per square yard.

## 2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

*Federal Specifications:*

CCC–T–191—Textile Test Methods.

PPP–P–51 — Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks and Synthetics).

*Federal Standards:*

Fed. Std. No. 4.—Glossary of Fabric Imperfections.

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

*Military Standard:*

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.



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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

**3.1 Standard sample.**—When specified, the cloth shall be equal to the standard sample in all respects not specifically covered by this specification (see 6.3).

**3.2 Preproduction sample approval.**—When specified, before production is commenced, the contractor shall submit for approval to the contracting officer, a 2-yard full width sample of the cloth he proposes to furnish (see 6.2).

**3.3 Material.**

**3.3.1** *Cotton.*—The cotton shall be of a suitable length of staple and grade to meet the requirements of this specification.

**3.3.2** *Yarn.*—The cotton used in the manufacture of the yarn shall be thoroughly cleaned, well carded, drawn, and evenly spun into singles yarn. The yarn shall be carded or combed as specified in table I. (See 6.7 for civilian agencies.)

**3.4 Color.**—The color shall be as specified, and shall be produced by the use of vat dyes to match an approved shade standard or shall be produced by the use of such colorants and/or such methods as may be specified by the contracting officer (see 6.2).

**3.4.1** *Matching.*—The color shall match the approved standard shade under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard shade under incandescent lamplight at 2800° Kelvin.

**3.4.2** *Colorfastness.*

**3.4.2.1** *Standard sample.*—The dyed cloth shall show colorfastness to laundering, bleaching, perspiration, crocking and light equal to or better than the standard sample, when tested as specified in 4.3.

**3.4.2.1.1** *Adjective rating (except Air Force).*—When no standard sample is available, or when the standard sample is not a reference for colorfastness, the dyed cloth, unless otherwise specified, shall show good colorfastness to laundering, bleaching, perspiration, crocking and light when tested as specified in 4.3 (see 6.2).

**3.4.2.1.2** *Adjeetive rating (Air Force).*—When no standard sample has been established, or when the standard sample is not a reference for colorfastness properties, the colorfastness requirements shall be specified by the contracting officer.

**3.5 Physical requirements.**—The finished cloth shall conform to the requirements shown in table I (see 6.5).

**3.5.1** *Width.*—The width shall be as specified and shall be the minimum acceptable width exclusive of selvages (see 6.5).

**3.6 Finish.** — The cloth shall be closely singed, desized, bleached and mercerized, and shall have a clear lustrous finish. Residual sizing, finishing or other nonfibrous material shall not exceed two percent when tested as specified in 4.3.

**3.7 Shrinkage.**—The cloth shall be preshrunk and shall not shrink more than one percent in either the direction of the warp or of the filling when tested as specified in 4.3. The preshrinking process used shall not be identified on the cloth, the ticket, or on the package.

**3.8 Length.**

**3.8.1** *Military.*—Unless otherwise specified, the cloth shall be in one continuous piece of not less than 50 yards.



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TABLE I.—*Physical requirements*

| Type | Yarn | Weight per square yard (minimum) | Yarns per inch (minimum) | | Breaking strength (grab method) (minimum) | | Weave |
|---|---|---|---|---|---|---|---|
| | | | Warp | Filling | Warp | Filling | |
| | | *Ounces* | | | *Pounds* | *Pounds* | |
| I | Combed | 3.5 | 148 | 74 | 74 | 28 | Plain |
| II | Combed | 3.2 | 140 | 56 | 65 | 25 | Plain |
| III | Combed | 3.2 | 132 | 64 | 75 | 30 | Plain |
| IV | Carded | 3.2 | 116 | 56 | 60 | 25 | Plain |
| V | Carded | 3.2 | 102 | 56 | 55 | 25 | Plain |
| VI | Carded | 2.8 | 82 | 56 | 30 | 25 | Plain |

3.8.2 *Nonmilitary.* — Unless otherwise specified, the cloth shall be in one continuous piece of not less than 45 yards and when furnished in bolts, not more than 55 yards.

**3.9 Tickets.**—Each piece shall have a ticket attached to the selvage at the inner end of the cloth with not finer than 5-ply cotton string doubled to not less than 8 inches long. The ticket shall be made of paper stock of not less than 0.020 inch in thickness and the color shall be white or light in intensity to permit easy reading of printed or penciled markings. The finish on the paper stock shall be suitable for pencil markings. The ticket shall have a reinforced eyelet for attaching the tying cord and shall be legibly printed with water insoluble ink with the following information, except when otherwise specified by civilian government agencies, and sufficient blank space shall be provided for the inspectors stamp.

Nomenclature
Type_____ Color_____ Width_____
Stock number
Contractor's name
Contract number and date
Contractor's piece number
Specification number and date
Name of contracting agency
Gross yards
Net yards

**3.10 Workmanship.**—The cloth shall conform to the quality and grade of product established by this specification, and the occurrence of defects shall not exceed the levels set by the applicable acceptable quality levels.

## 4. SAMPLING, INSPECTION, AND TEST PROCEDURES

**4.1 Nonmilitary.** — Nonmilitary agencies may use sampling, inspection, and test procedures outlined under paragraph 4.2 when desirable. When procedures outlined under 4.2 are not deemed suitable, the procedures specified under nonmilitary paragraphs shall be followed.

4.1.1 *Sampling.*

4.1.1.1 *Inspection lot.* — Unless otherwise specified in the invitation for bids, all cloth of one type submitted at one time shall be considered a lot for purposes of inspection and tests.

4.1.1.2 *Sampling for inspection.*—A random sample of cloth shall be taken from each lot in accordance with table II.



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TABLE II.—*Sampling for inspection*

| Number of yards in lot | Number of rolls or bolts in the inspection sample | Number of yards in the inspection sample [1] | Acceptance number—total, defective | Rejection number—total, defective | Acceptance number—major, defective | Rejection number—major, defective |
|---|---|---|---|---|---|---|
| 300 and under | 2 | 35 | 5 | 6 | 1 | 2 |
| 301 to 500 | 3 | 50 | 6 | 7 | 2 | 3 |
| 501 to 800 | 5 | 75 | 9 | 10 | 3 | 4 |
| 801 to 1,300 | 7 | 110 | 12 | 13 | 4 | 5 |
| 1,301 to 3,200 | 10 | 150 | 17 | 18 | 5 | 6 |
| 3,201 to 8,000 | 15 | 225 | 24 | 25 | 8 | 9 |
| 8,001 to 22,000 | 20 | 300 | 32 | 33 | 10 | 11 |
| 22,001 to 110,000 | 30 | 450 | 43 | 44 | 14 | 15 |
| 110,001 to 550,000 | 40 | 750 | 68 | 69 | 20 | 21 |
| 550,001 and over | 60 | 1500 | 124 | 125 | 35 | 36 |

[1] Yards in the inspection sample shall be divided equally among sample rolls of cloth and that number of yards shall be unrolled from each sample roll.

4.1.1.3 *Sampling for tests.*—Random samples for tests shall be taken from each inspection lot in accordance with table III.

TABLE III.—*Sampling for tests*

| Number of rolls or bolts in lot | Number of rolls or bolts in test sample[1] | Acceptance number (defective) | Rejection number (defective) |
|---|---|---|---|
| 8 and under | 2 | 0 | 1 |
| 9 to 15 | 3 | 0 | 1 |
| 16 to 40 | 4 | 0 | 1 |
| 41 and over | 5 | 0 | 1 |

[1] One unit of product (test sample) at least 2 yards long and the full width of the cloth shall be taken from each sample roll.

4.1.2 *Preproduction sample inspection.*—When a preproduction sample is required it shall be inspected and tested as specified in 4.2.2.1.

4.1.3 *Yard-by-yard inspection.*—Each of the samples selected in accordance with table II, shall be visually inspected and defects classified in accordance with 4.2.3.2. The maximum number of defects acceptable shall be as specified in table II.

4.1.4 *Overall inspection and inspection for length.*—Overall inspection and inspection for length shall be as specified in 4.2.3.3 and 4.2.3.4, respectively. Sample size and the maximum number of defects acceptable shall be as specified in table IV.



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4.1.5 *Tests.*—Samples selected for tests in accordance with table III shall be tested in accordance with the applicable methods of Federal Specification CCC–T–191 as listed in table V of this specification. The physical and chemical values specified in section 3 of this specification apply to the average of the determinations made on a unit of product for test purposes as specified in applicable test methods. The unit of product shall be 2 continuous yards, full width of finished cloth. The number of defects acceptable shall be as specified in table III.

4.1.6 *Compliance criteria.*

4.1.6.1 *Inspection.*—Any piece in the sample for inspection, failing to conform to the requirements for which the inspection was made shall be considered defective. If the number of defective pieces in the sample exceeds the acceptance number for that sample the entire lot from which the sample was taken shall be considered to have failed to meet the requirements of this specification.

4.1.6.2 *Tests.*—Any piece in the sample for tests, failing to conform to the requirements for which tests were made shall be considered defective. If the number of defective pieces in the sample exceeds the acceptance number for that sample, the entire lot from which the sample was taken shall be considered to have failed to meet the requirements of this specification.

**4.2 Military.**

4.2.1 *General.*—The quality assurance provisions contained in this section are applicable to inspection. Quality assurance provisions for components or materials described shall be in accordance with this specification and with subsidiary specifications referred to herein to the extent applicable, except that this specification shall govern in the event of conflict.

4.2.2 *Preacceptance inspection.*

4.2.2.1 *Preproduction sample inspection.*—The preproduction sample, if submitted in accordance with 3.2 shall be visually inspected for appearance, color and finish. The sample shall be tested for physical and chemical properties in accordance with the methods specified in 4.3.

4.2.3 *Inspection for acceptance.*—Inspection shall be in accordance with the provisions of Military Standard MIL–STD–105 except where otherwise indicated hereinafter.

4.2.3.1 *Examination of the end item.*—The cloth shall be examined to determine conformance with the requirements of this specification. Defects found during this examination shall be classified in accordance with 4.2.3.2, 4.2.3.3 and 4.2.3.4.

4.2.3.2 *Yard-by-yard examination.* — The defects listed herein shall be counted regardless of their proximity one to another, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The unit of product for this examination shall be one yard. The acceptable quality levels for this examination shall be 2.5 major and 10.0 total defects (major and minor combined) per 100 units (yards). The sample size shall be based upon inspection level III, table III of Military Standard MIL–STD–105. The lot size shall be expressed in units of 1 yard each. The number of pieces from which the sample is to be selected shall be in accordance with table IV. An approximate equal number of yards shall be examined from each piece in the sample.



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| Glossary No. Fed. Std. No. 4 | Defect | Classification | |
|---|---|---|---|
| | | Major | Minor |
| 90, 91, 92 | Hole, cut or tear, three or more threads ruptured at adjoining points | X | |
| 78, 79, 89 | Smash, bruise, tender or weak spot—any | X | |
| 11, 33 | Broken or missing yarns— | | |
| | Two or more contiguous, more than ½ inch missing | X | |
| | Single, 2 inch or more missing | | X |
| 9 | Fine filling bar or light place | X | |
| 8, 25 | Fine, uneven or light yarns—clearly noticeable | | X |
| 1, 2, 26 | Coarse filling bar or heavy place | X | |
| | Coarse yarns, clearly noticeable | | X |
| 5, 6, 27, 77 | Kink, slub or gout— | | |
| | Over 1/16 inch in thickness | X | |
| | 1/16 inch or less in thickness, clearly noticeable | | X |
| 7, 45 | Surface knot or fuzz ball—more than ⅛ inch in diameter | | X |
| 3, 5 | Jerked in filling, slough off— | | |
| | Over 1/16 inch in thickness, regardless of length | X | |
| | 1/16 inch or less in thickness and more than 9 inches in length | X | |
| | 1/16 inch or less in thickness, but not over 9 inches in length—clearly noticeable | | X |
| 17, 35 | Open place, or hitchback— | | |
| | Resulting in a weak spot | X | |
| | Not resulting in a weak spot, clearly noticeable | | X |
| 79 | Abrasion mark—clearly noticeable fuzziness but not resulting in a weak spot | | X |
| 4, 10 | Mispick, two or more picks in same shed | X | |
| 13 | Floats or harness skips— | | |
| | Multiple ¼ inch or more in combined warp and filling directions | X | |
| | Multiple less than ¼ inch in combined warp and filling directions, or single two inches or more | | X |
| 16, 37, 38, 40, 41 | Tight or slack yarn, slack or tight twist yarn, wrong draw, reed mark—clearly noticeable | | X |
| 42, 76 | Temple mark—clearly noticeable | | X |
| 49, 50 | Selvage tight—resulting in pronounced slackness or waviness in cloth | | X |
| 51 | Selvage scalloped—indentation exceeding ½ inch | | X |
| 52 | Selvage slack—resulting in pronounced waviness | | X |
| 69, 70, 74, 75, 76 | Spot, stain or streak—clearly noticeable | X | |
| 14, 57 | Shade bar, mixed filling, finishing mark | X | |



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4.2.3.3 *Over-all examination.*—Each defect listed shall be counted no more than once in each piece examined. The unit of product for this examination shall be one piece. The sample size (number of pieces selected as sample) for this examination and the number of defects acceptable shall be as shown in table IV.

*Defects*

Width less than specified.
Uneven weaving, clearly noticeable.
Baggy, ridgy, or wavy cloth—clearly noticeable, will not flatten under normal pressure.
Selvage curled, folded, rolled or slack—extending intermittently or continuously over more than half the length of the piece.
Off shade, shaded side to side, side to center, end to end, or throughout the piece.
Mottled, cloudy or poor penetration.
Objectionable odor, overall uncleanness.

TABLE IV.

| Lot size in yards | Sample size in pieces | Acceptance number |
|---|---|---|
| Up to 1,300[1] | 3 | 0 |
| 1,301 to 3,200 | 5 | 0 |
| 3,201 to 8,000 | 7 | 0 |
| 8,001 to 22,000 | 10 | 0 |
| 22,001 to 110,000 | 15 | 0 |
| 110,001 to 550,000 | 25 | 1 |
| 550,001 and over | 50 | 2 |

[1] If a lot contains fewer than three pieces, each piece in the lot shall be examined.

4.2.3.4 *Examination for length.*

4.2.3.4.1 *Examination for length of individual pieces.*—The pieces shall be examined for gross length. Any gross length found to be less than the specified minimum length, or any gross length found to be more than 2 yards below the gross length marked on the piece ticket, shall be considered as a defect with respect to length. The unit of product for this examination shall be one piece. The number of pieces selected as sample (sample size) for this examination and the number of defects acceptable shall be in accordance with table IV.

4.2.3.4.2 *Total yardage in sample.*—The lot shall be unacceptable if the total of the actual gross length of pieces in the sample is less than the total of the gross lengths marked on piece tickets. The pieces examined shall be those selected for examination of individual pieces.

**4.3 Testing of the end item.**—The methods of testing specified in Federal Specification CCC–T–191, wherever applicable, and as listed in table V shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test method. The unit of product shall be three continuous yards, full width of the finished fabric. For each characteristic, the acceptable quality level shall be 4.0 test failures per hundred units (yards), and the inspection level shall be level L–1, table VII, of Appendix to Military Standard MIL–STD–105. The lot size shall be expressed in terms of 1 yard each.

## 5. PREPARATION FOR DELIVERY

(For definitions and levels, see 5.4).

**5.1 Packaging.**

5.1.1 *Levels A and B.*—The cloth shall be packaged in accordance with the applicable requirements for class 3 fabrics of Federal Specification PPP–P–51 except that when specified by civilian government agencies, material purchased in bolt form shall be wrapped in waterproof or other paper as specified in the contract or order.

5.1.2 *Level C.*—The cloth shall be packaged in accordance with the manufacturer's normal practice, except that bolts, when procured for civilian agencies, shall be com-



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TABLE V.—*Test methods*

| Characteristic | Requirement reference | Test method |
|---|---|---|
| Weave | Table I | Visual |
| Yarns per inch | Table I | 5050 |
| Weight, ounces per square yard | Table I | 5041 |
| Breaking strength | Table I | 5100 |
| Colorfastness to: | | |
| Light[1] | 3.4.2 | 5660 |
| Chlorine bleaching | 3.4.2 | 5600 |
| Perspiration[2] | 3.4.2 | 5680 |
| Crocking | 3.4.2 | 5650 |
| Laundering | 3.4.2 | 5610 |
| Shrinkage in laundering | 3.7 | 5550 |
| Nonfibrous materials[3] | 3.6 | 2611 |
| Mercerization | 3.6 | Microscopic examination for swelling of fibers |

[1] In the event of a dispute resulting from the test with method 5660, or as a result of suspected anomalous behavior of certain dye types or formulations, the contracting officer may authorize the exposure to natural light in accordance with method 5662.

[2] The transfer cloth shall consist of a warp of Orlon 81 and 6-fiber filling of equal bars of wool, spun nylon, spun rayon, spun silk, spun acetate and bleached cotton. Each 6-fiber repeat shall measure approximately 2 inches and shall be separated by a waste filling stripe.

[3] The nonfibrous content shall be determined in accordance with method specified and shall be calculated from the following formula:

$$\text{Nonfibrous content, percent} = \frac{O - S}{O} \times 100$$

pletely wrapped in kraft paper of suitable weight to afford protection from soiling and similar damages. Wrapping shall encircle the bolt at least one and one-half times and shall be sufficiently wide to fold over at each end to protect ends of bolt. Folded ends shall be secured by use of gummed tape. The bolt shall be wrapped so as to leave the identification ticket clearly exposed on the outside of the package.

**5.2 Packing.**

**5.2.1** *Levels A and B.*—The cloth shall be packed in accordance with the applicable requirements of Federal Specification PPP–P–51.

**5.2.2** *Level C.* — The cloth, packaged as specified in 5.1.2, shall be packed in a manner acceptable to common carriers for safe transportation to point of destination specified in shipping instructions at the lowest transportation rate for such supplies, except that for civilian government agencies the material furnished in bolt form shall be packed one bolt wide and four bolts high in a shipping container.

**5.3 Marking.**—Shipments shall be marked in accordance with the requirements of Federal Specification PPP–P–51.

5.4 Federal Standard No. 102 shall be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

## 6. NOTES

**6.1 Intended use.**—The cloth, cotton, broadcloth covered by this specification is intended for use in the manufacture of clothing items.

**6.2 Ordering data.** — Procurement documents should specify the following:

*a.* Title, number, and date of this specification.
*b.* Type of cloth required (see 1.2).
*c.* Length of cloth (see 3.8).
*d.* Color of cloth required (see 3.4).
*e.* Colorfastness, requirement and evaluation other than specified (see 3.4.2).
*f.* When standard sample is required (see 3.1).
*g.* When preproduction sample approval is required (see 3.2).
*h.* Width of cloth required (see 3.5.1).



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*i.* Both the level of packaging and the level of packing required (see 5.1, 5.2 and 6.4).

*j.* Packaging requirements for levels A and B when material is purchased in bolt form — civilian agencies (5.1.1).

**6.3 Standard sample.**—For access to standard, address the procuring office issuing the invitation for bids.

6.4 In the preparation of contracts or orders, it should be noted that various options, choices, and alternatives, as indicated in Federal Specification PPP–P–51, may be exercised in the preparation for delivery of this commodity.

**6.5 Commercial grey designation.** — It is suggested that cloth of the following commercial grey designations (or prorata) be used to produce cloth meeting the requirements of this specification.

*Commercial grey designation (suggested)*

| Type | Width | Yarns per inch | Weight yards per pound |
|---|---|---|---|
| I | 37½ | 144 x 76 | 4.15 |
| II | 37 | 136 x 60 | 4.00 |
| III | 37 | 128 x 68 | 4.00/4.25 |
| IV | 37 | 112 x 60 | 3.85 |
| V | 37 | 100 x 58 | 4.00 |
| VI | 40 | 80 x 60 | 4.50 |

**6.6 Transportation description.**—The transportation description applicable to this item is:

Cotton piece goods, in the original piece, finished, not otherwise indexed by name.
Carload minimum weight, any quantity.
Motor volume minimum weight, any quantity.

**6.7 Procurement by civilian agencies.** — Broadcloth procured by civilian government agencies under this specification may contain yarns in excess of single ply providing the broadcloth conforms with all other requirements of this specification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

**MILITARY INTERESTS:**
**Army—M Q**
**Navy—S**
**Air Force.**



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